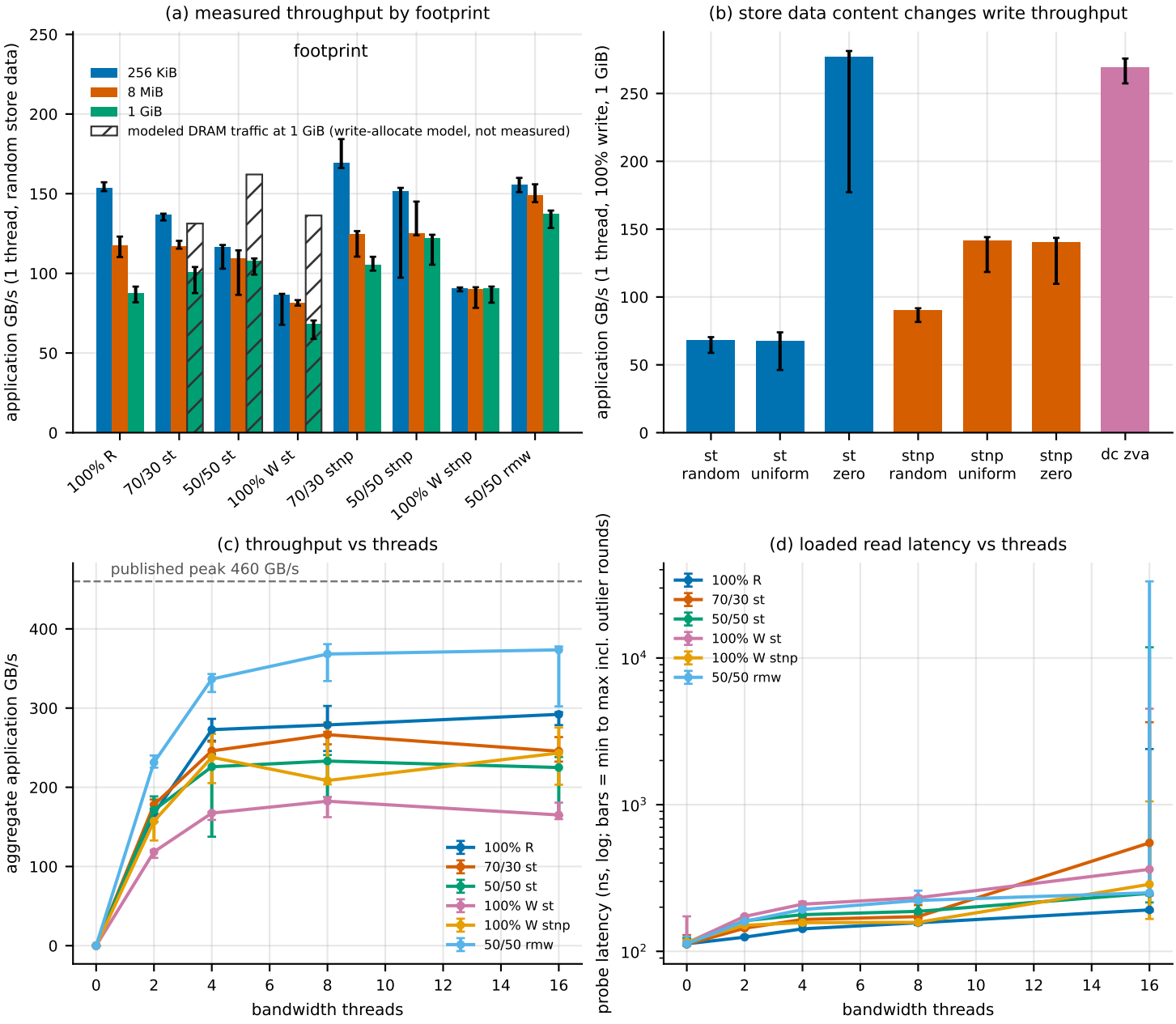
(a) measured throughput by footprint
footprint
256 KiB
8 MiB
1 GiB
modeled DRAM traffic at 1 GiB (write-allocate model, not measured)
application GB/s (1 thread, random store data)
100% R
70/30 st
50/50 st
100% W st
70/30 stnp
50/50 stnp
100% W stnp
50/50 rmw
(b) store data content changes write throughput
application GB/s (1 thread, 100% write, 1 GiB)
st random
st uniform
st zero
stnp random
stnp uniform
stnp zero
dc zva
(c) throughput vs threads
published peak 460 GB/s
aggregate application GB/s
bandwidth threads
(d) loaded read latency vs threads
probe latency (ns, log; bars = min to max incl. outlier rounds)
bandwidth threads